Loan Details Analysis and Performance Dashboard

1. Project Overview

This project focuses on analysing loan data to understand loan distribution, approval patterns, and financial trends. The dataset contains information such as loan amount, interest rate, loan term, loan status, state, and purpose of the loan.

2.Project Objective

The main objectives of this project are:

- To analyse loan distribution across different states and purposes.
- To evaluate loan approval and rejection patterns.
- To study the relationship between loan amount and interest rate.
- To identify trends in loan origination over time.
- To create an interactive dashboard using Power BI for better decision-making.

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2. Data Sources

- **Source Description and Timeline:** https://practicedatasets.online/lending-club-datasets and 2023-2026
- **Domain**: Banking and Financial Services

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3. Problem Statement

- To analyse loan distribution across different states and purposes to identify high-performing regions and major borrowing categories.
- To evaluate loan approval, rejection, and default patterns to understand overall loan performance.
- To examine the relationship between loan amount, interest rate, and loan term to identify financial risk factors.
- To study loan origination trends over time for better financial planning and decision-making.
- To develop an interactive Power BI dashboard to transform raw loan data into meaningful business insights.

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4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Loan ID	Text	Unique ID for Loan
Loan Amount	Currency	Amount of loan sanctioned
Interest Rate	Percentage	Percentage interest for loan amount

Loan Term (Months)	Number	Number of months allotted for loan repayment
Purpose	Text	Reason for loan application
Origination Date	Short Date	Date of Sanction
Loan Status	Text	Repayment condition of the loan
Monthly Payment	Currency	Amount paid every month
State	Text	State of the Borrower

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5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Excel)

Tasks Performed:

- **Data Cleaning & Transformation:**
- Standardized formats
- Created calculated fields

  1.Default count =IF(D2="Default",1,0)

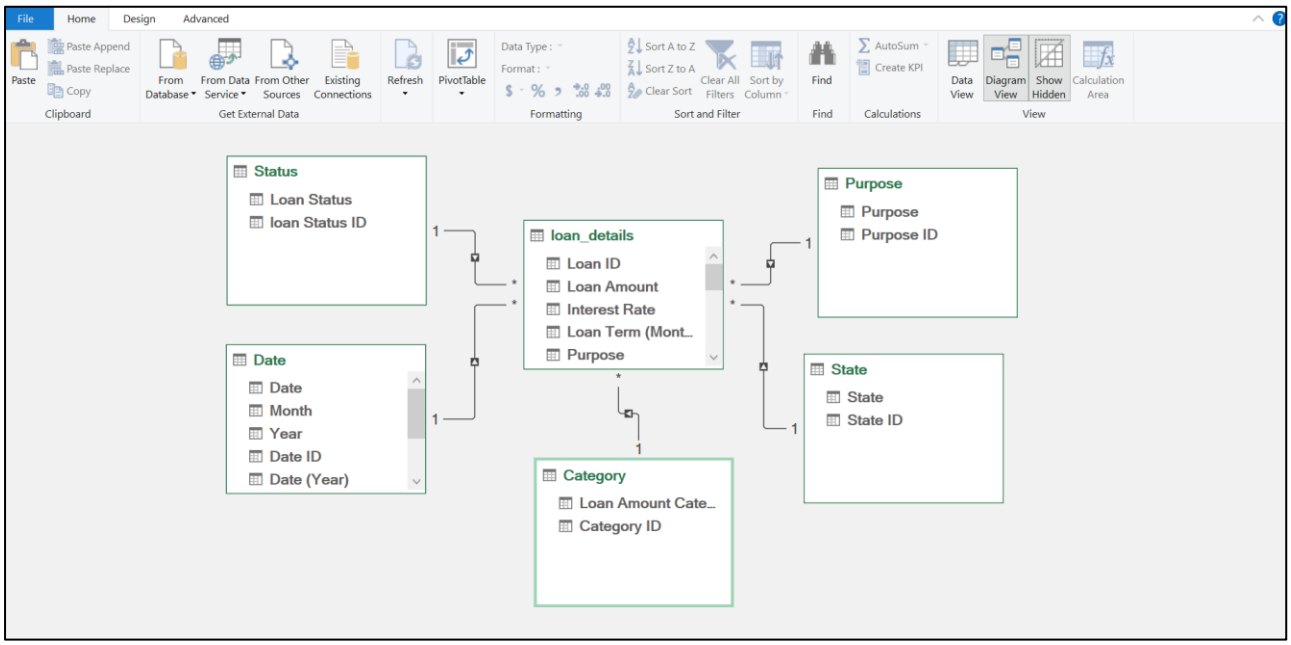
  2.Category = =IF([@[Loan Amount]]<20000, "Below Average", IF([@[Loan Amount]]<30000, "Average","Above Average"))
- Applied clean and Trim formulas to Purpose, Loan status and state columns.
- **Sorting:** Sorted date column from Newest to oldest.
- Convert the data into Fact and Dimension Table**.** Created 5 dimension tables.
  1. Dim State
  2. Dim Purpose
  3. Dim Status
  4. Dim Date
  5. Dim Category and connected fact and Dimension tables by Data modelling.
- Extracted State ID, Purpose ID, Status ID, Date ID and Category ID from dimension tables to fact tables using Vlookup function.

  E.g.,

  =VLOOKUP([@[Loan Amount Category]],Category[#All],2,FALSE)
- Likewise for all other columns.
- Kept only Numerical columns in fact table and removed the categorical columns for which dimension tables created.

- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary

SCREENSHOT OF DATA MODELLING IN EXCEL

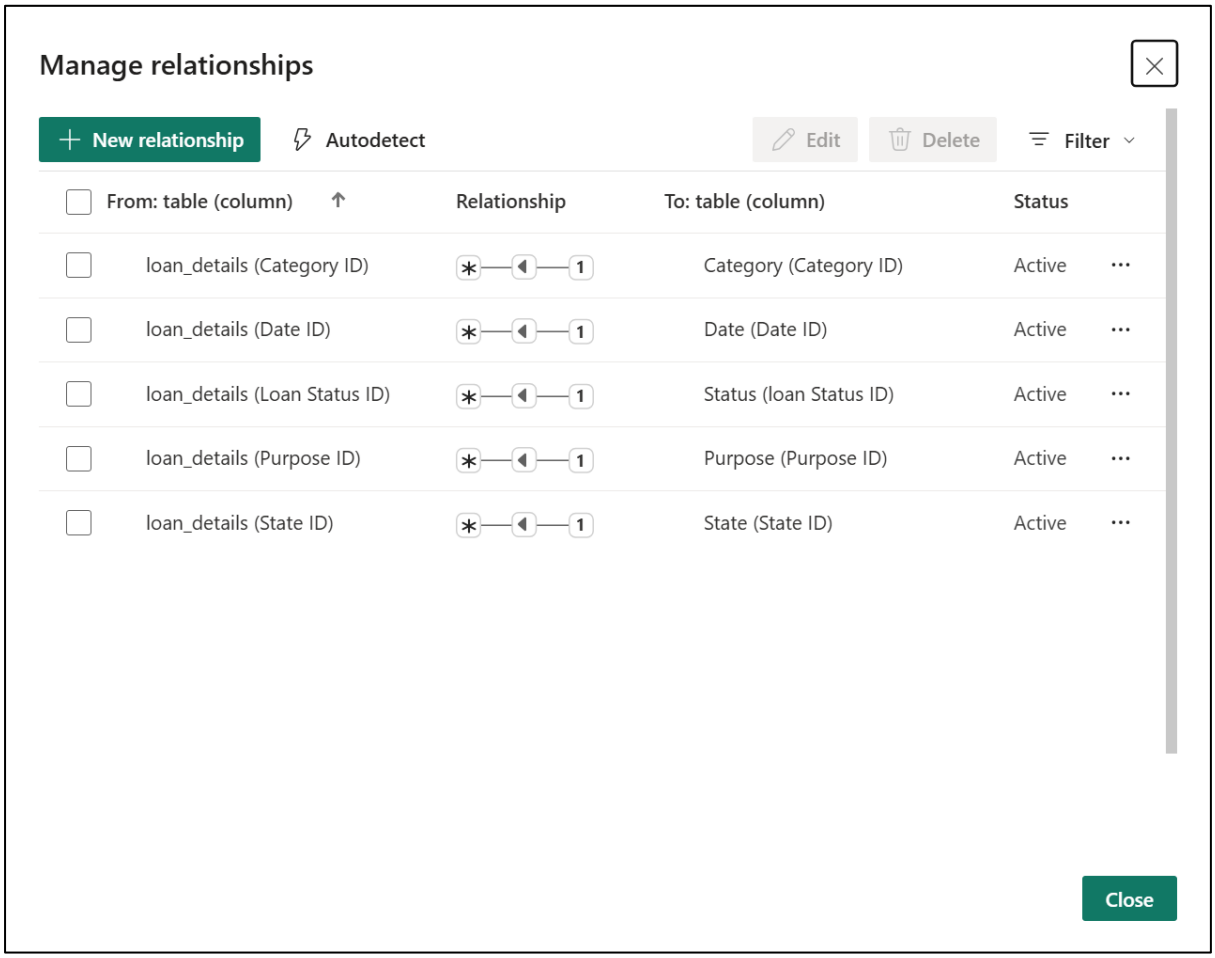


- Added new sheet named Descriptive statistics and calculated the Average and Sum of Loan Amount, Interest rate, Loan term, Default Count and Monthly payment using Average and Sum functions.
- Identified Mean, Median, Mode, Standard Deviation, Kurtosis, Skewness, Range, Maximum, Minimum, Sum, Count and Sample variance for Loan Amount, Interest rate Loan term and monthly payment columns using Analysis Toolpak method.
- **Pivot Tables:** Generated Pivot Tables for data summarisation and initial insights. Totally eight pivot charts created to identify various Trends, Patterns and Analysis.

7. Data Modelling and DAX (Power BI)

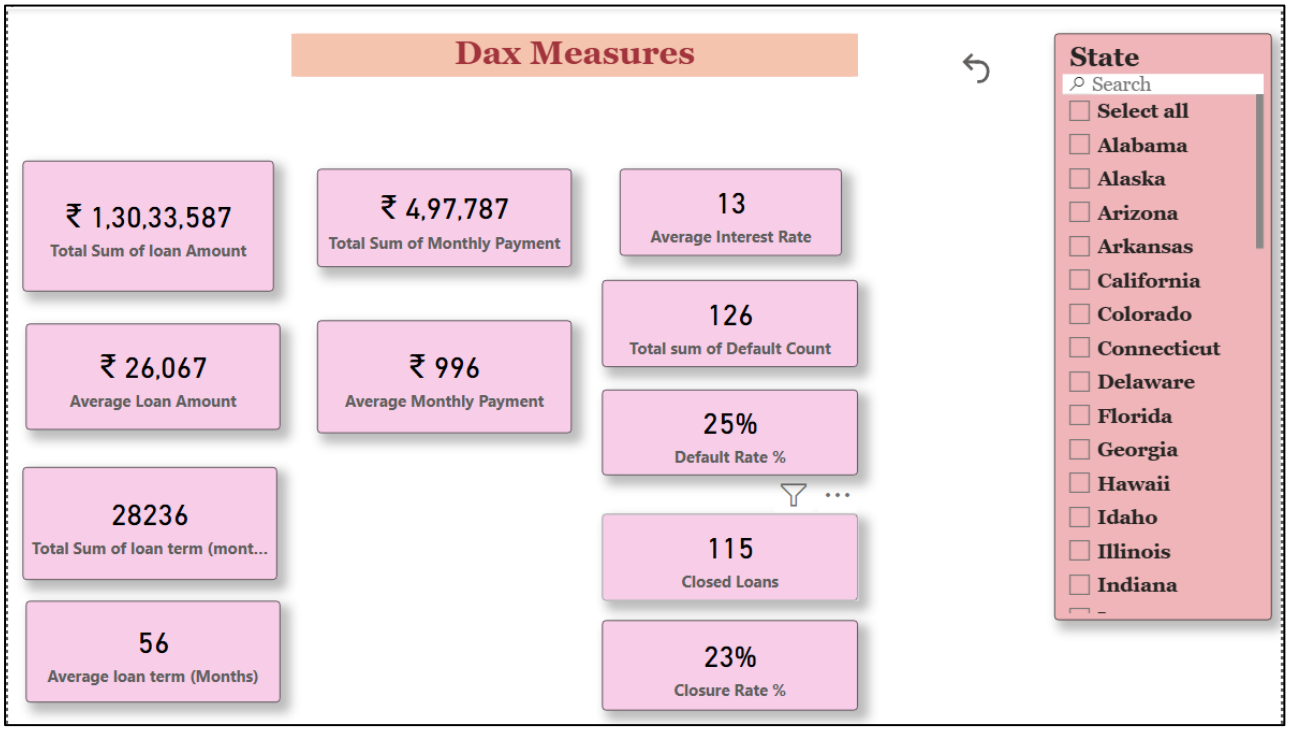
- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary.

SCREENSHOT OF DATA MODELLING



- **DAX Measures:** Implemented DAX formulas for key metrics, such as

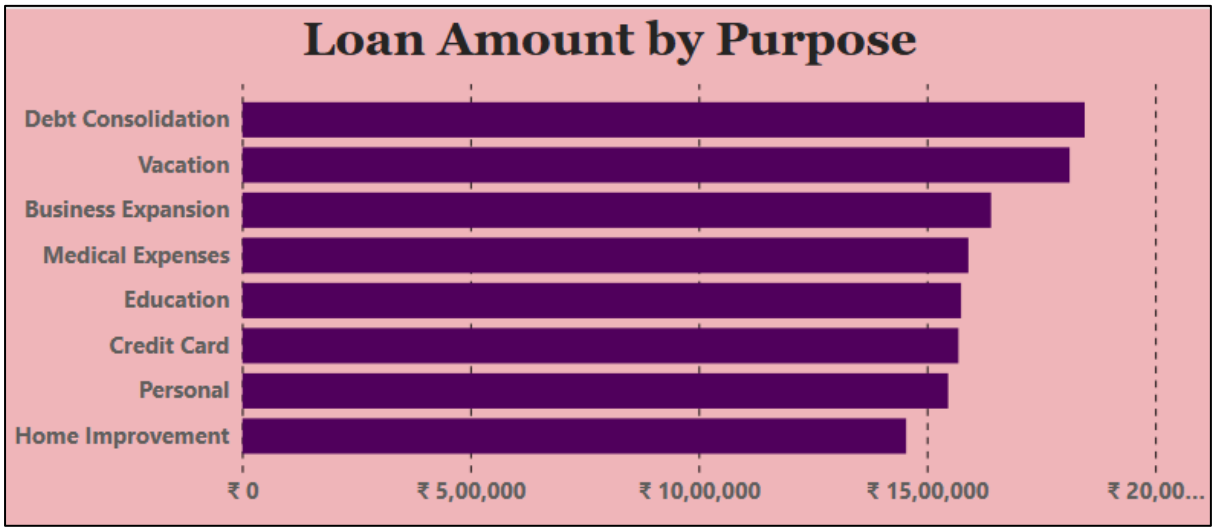
1. **Total Sum of loan Amount = Sum(loan_details[Loan Amount])**
2. Average Loan Amount = Average(loan_details[Loan Amount])
3. Total Sum of loan term (months) = Sum(loan_details[Loan Term (Months)])
4. Average loan term (Months) = Average(loan_details[Loan Term (Months)])
5. Total Sum of Monthly Payment = Sum(loan_details[Monthly Payment])
6. Average Monthly Payment = Average(loan_details[Monthly Payment])
7. Average Interest Rate = Average(loan_details[Interest Rate])
8. Total sum of Default Count = sum(loan_details[Default count])
9. Default Rate % = DIVIDE ([Total sum of Default Count], COUNT (loan_details[Loan ID]), 0)
10. Closed Loans = Calculate(count(loan_details[Loan ID]), loan_details[Loan Status]= "Paid off")
11. Closure Rate % = DIVIDE([Closed Loans], COUNTROWS(loan_details), 0)



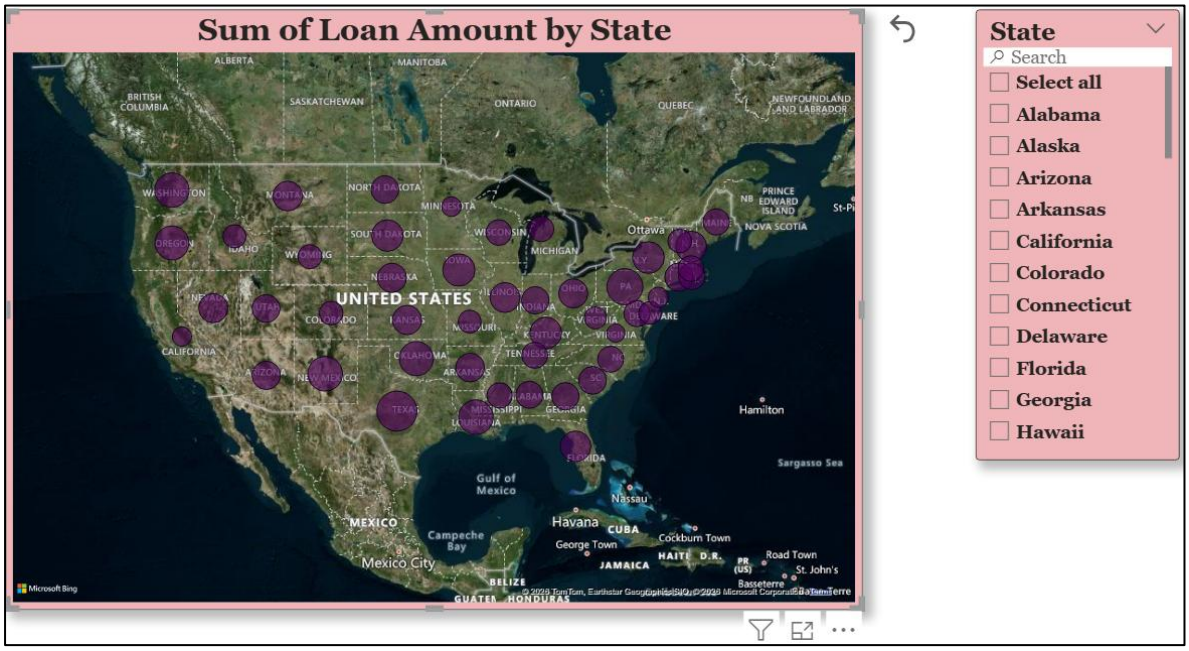
8. Analysis and Visualizations (Power BI)

Dashboard Features:

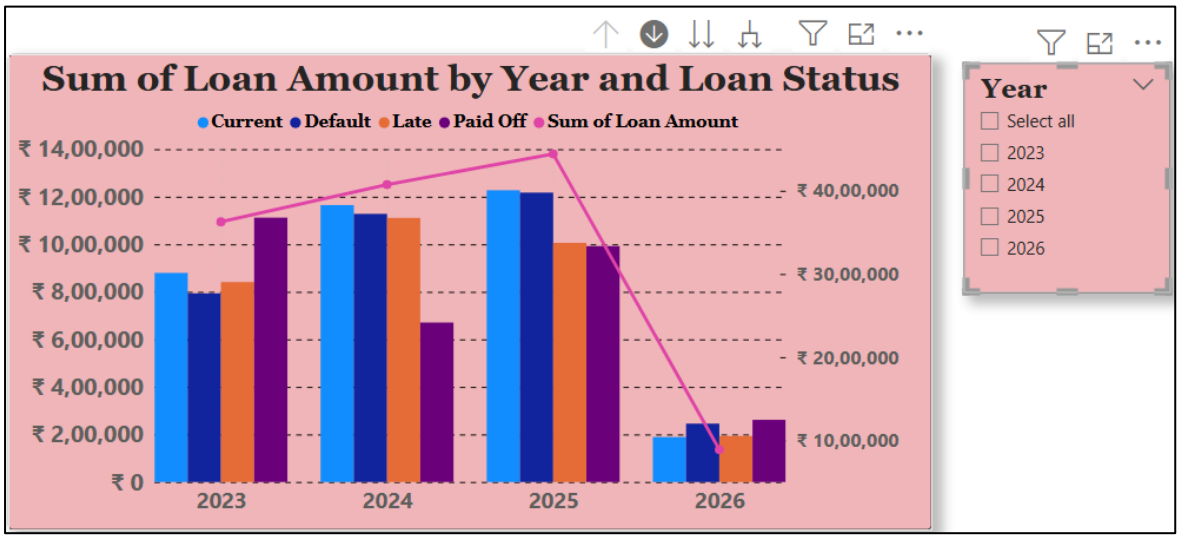
- **Multiple Visualizations based on problem statement:**



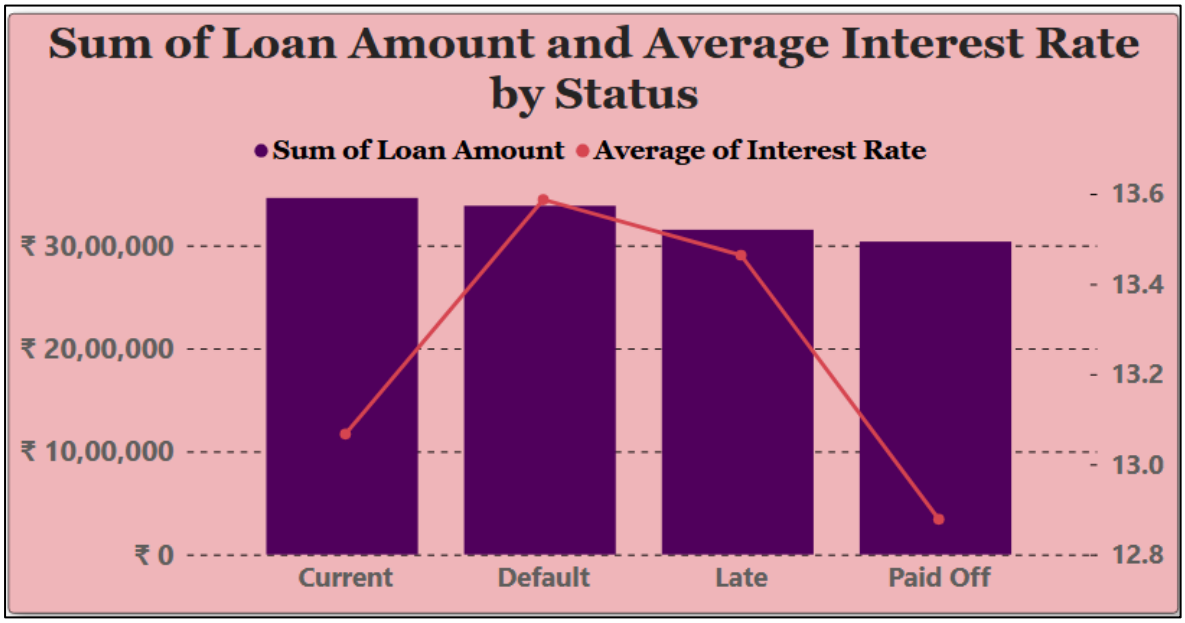
- ❖ Chart plotted to analyse the purpose for which maximum loan amount Sanctioned.
- ❖ By seeing the visual it clearly shows the Debt Consolidation and Vacation contributes the most and Home Improvement contributes least.
- ❖ It also varies for each State.



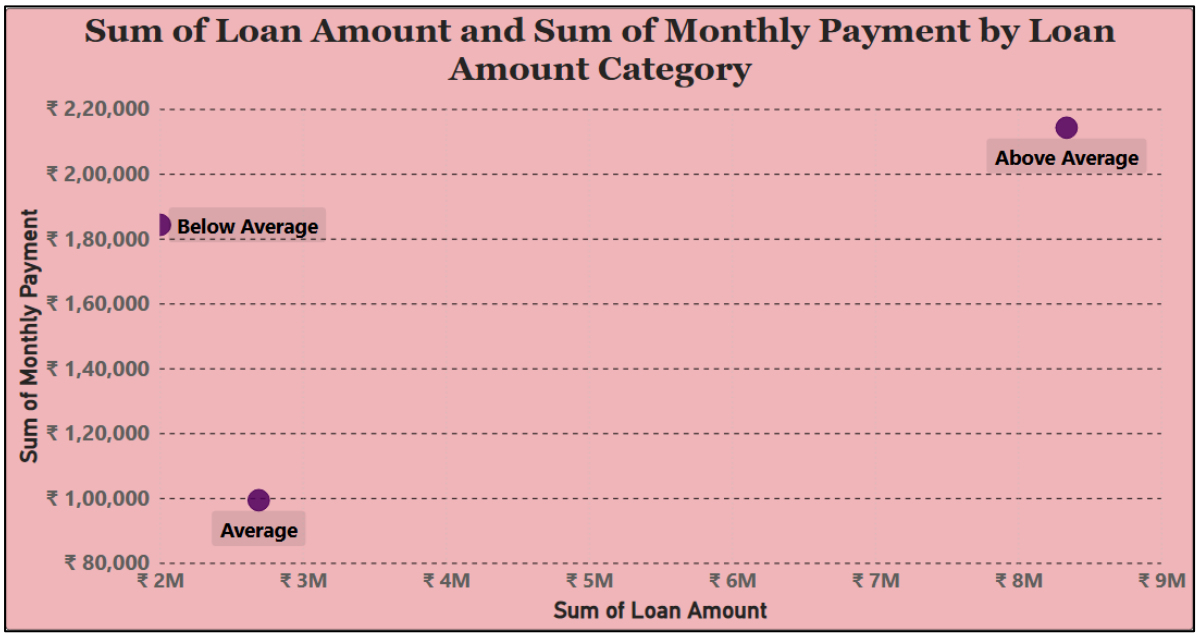
- Map visualization helps understanding the geographical view of Loan amount distribution across the country.
- State slicer also included to filter the state and Reset filter button to clear the filter.



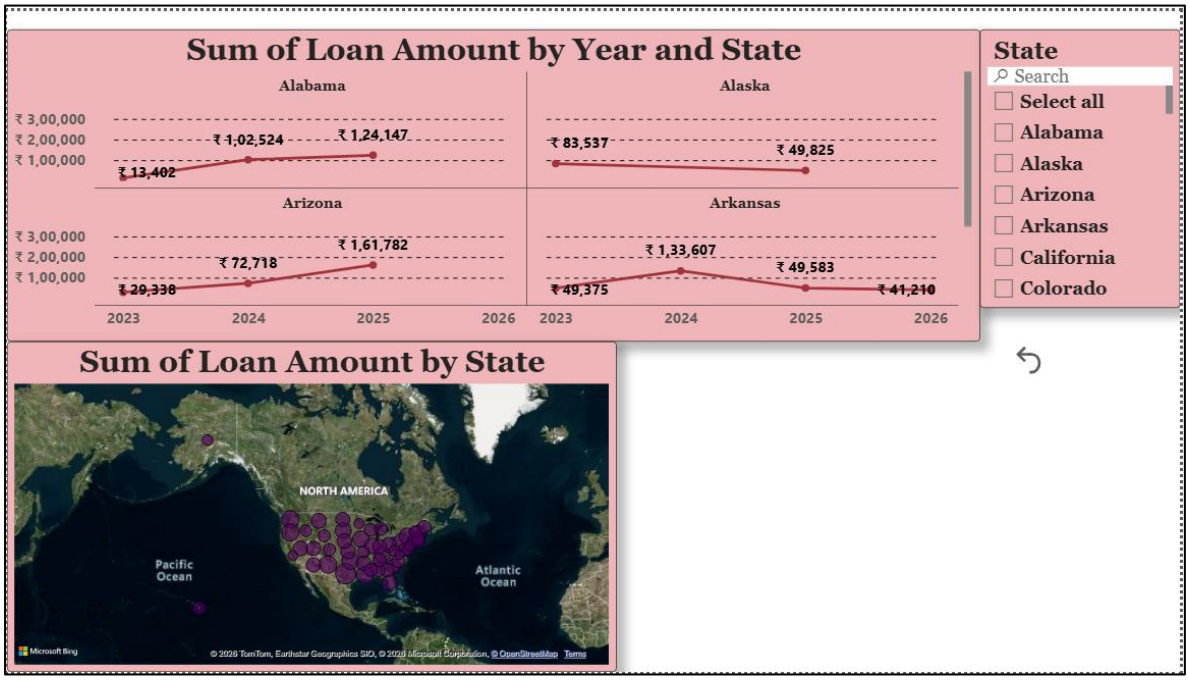
- This chart plotted to identify the trend of loan distribution across years and various status of the loans sanctioned.
- Year slicer included to filter the particular year. Also drill down option enabled to see months of the year.
- In the year 2025 only a greater number of loans sanctioned and more default loans but 2023 shows high paid off rate.



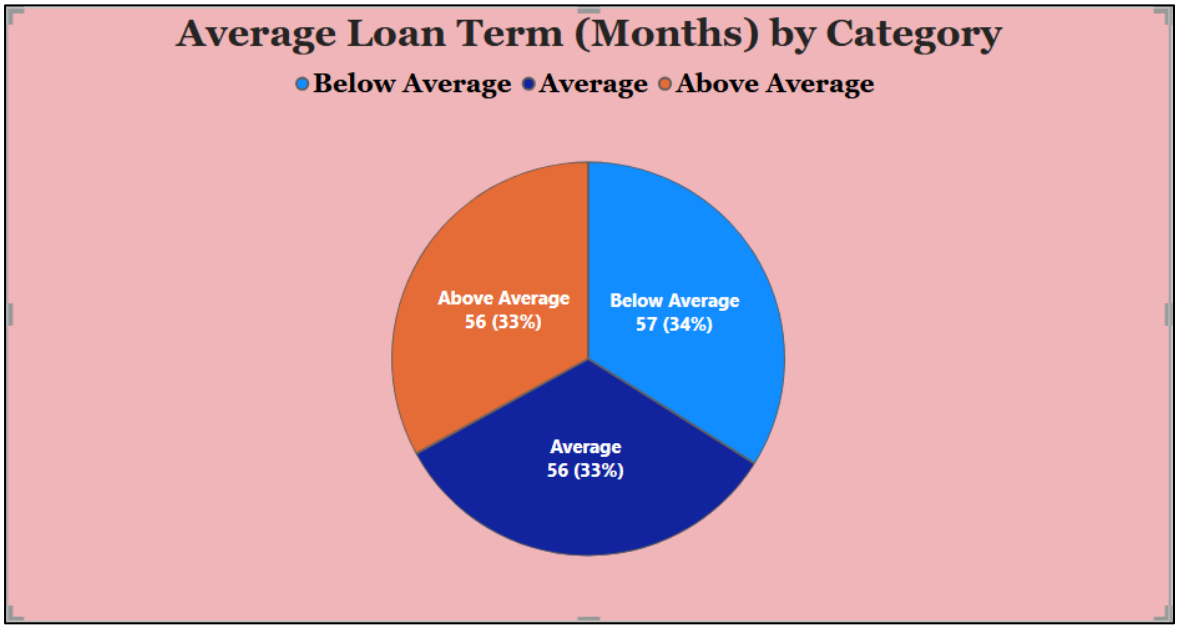
- This chart clearly shows the impact of interest rate in the loan status.
- **Default interest rate = 13.59, Paid off interest rate = 12.88**
- **Current Interest rate = 13.07, Late interest rate = 13.46**



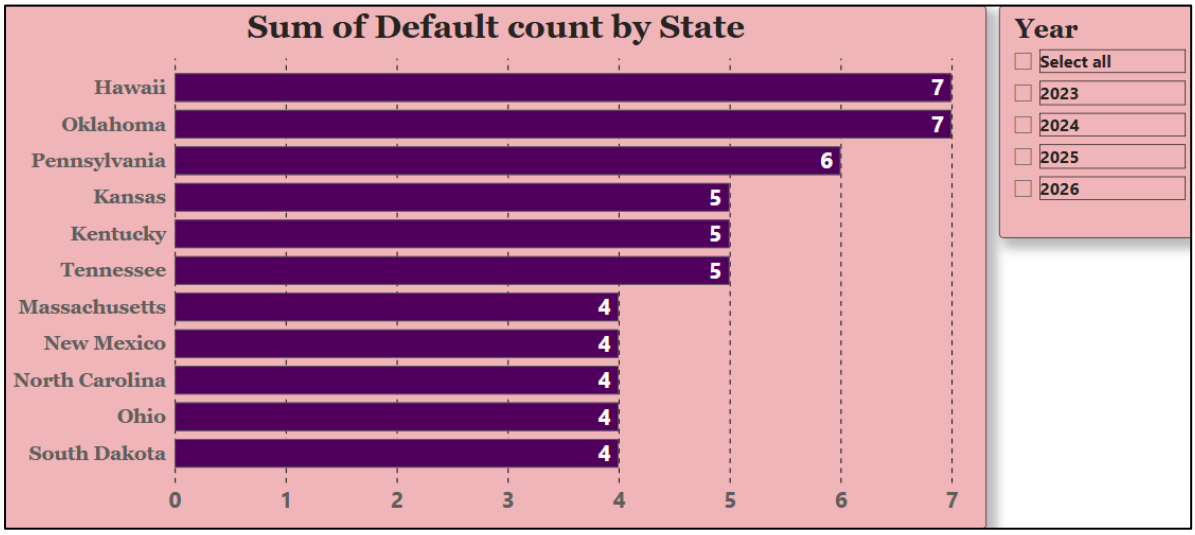
- Here below average loan amount shows high sum of monthly payment compared to Average loan amount.
- This visualization helps comparing the loan amount and monthly payment by category.



- This chart shows the trend by state. This helps in analysing the loan amount sanctioned to each state across the years and their pattern.
- State slicer also added to filter the particular state to be analysed.

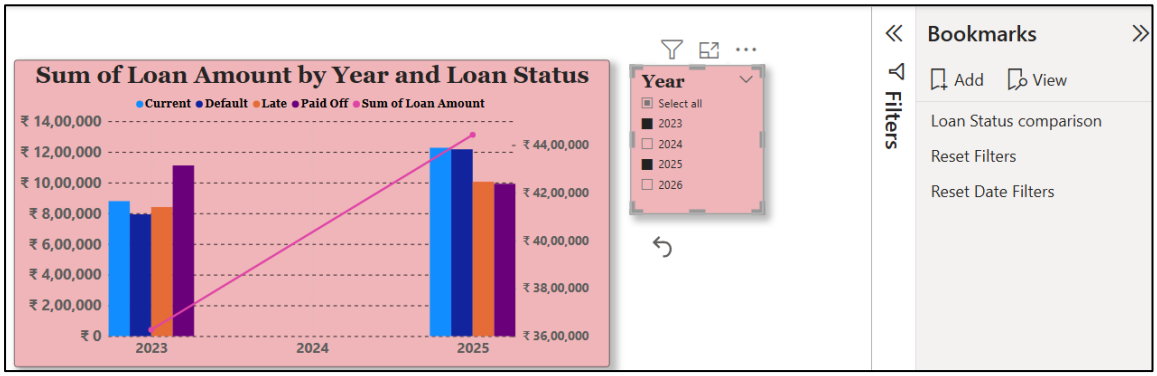


- This visualization helps in comparing the average loan terms by category.
- Same as monthly payment Below average category has greater loan term compared to Average and Above Average.



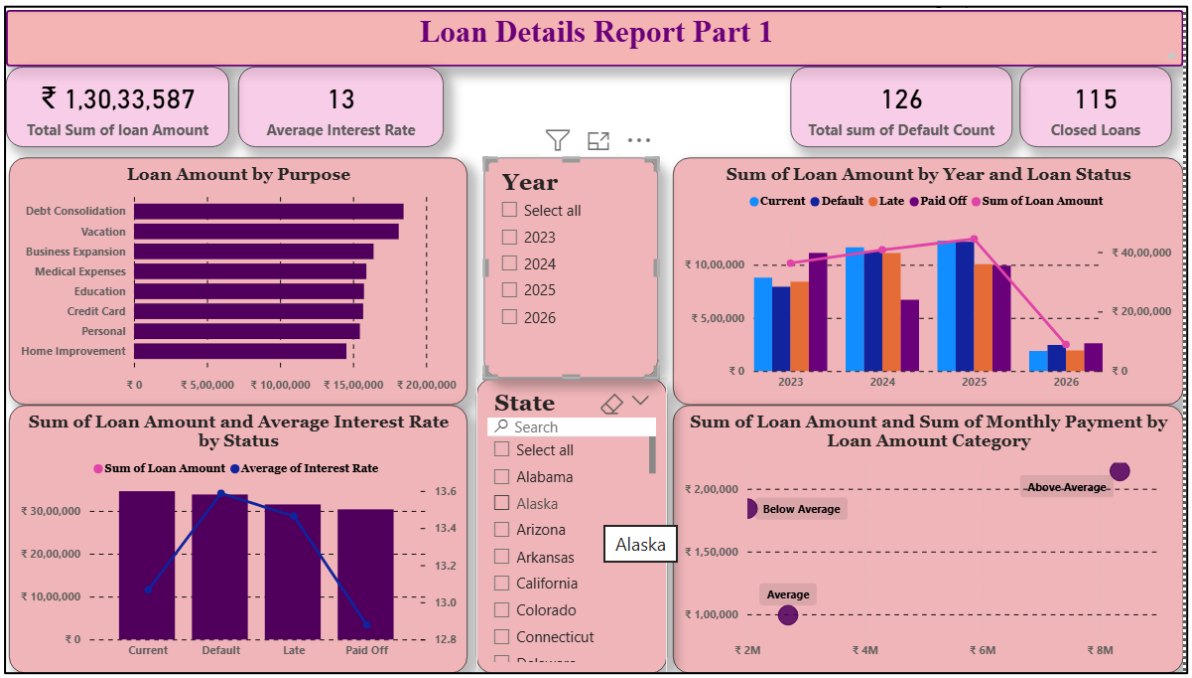
- Top 10 states filtered by sum of default count from 50 states.
- Year slicer applied to filter by year. This helps in analysing the Risk.

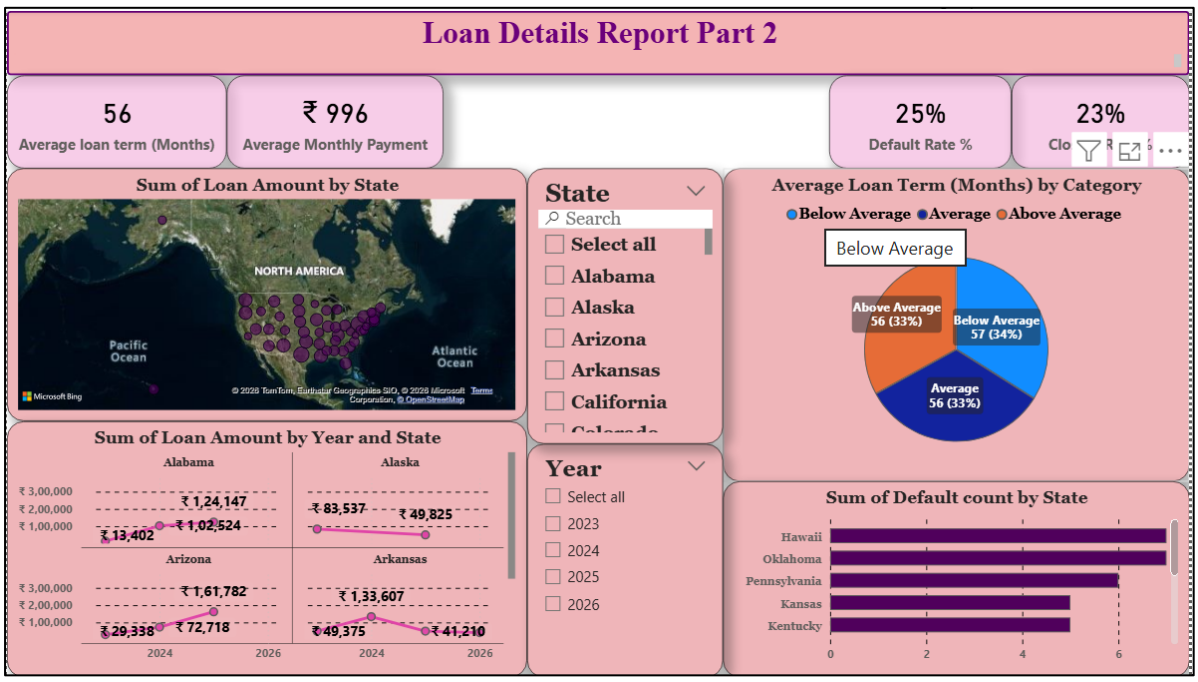
Bookmark



- Bookmark created to compare the loan amount by year and loan status.

REPORTS:





9. Insights

- **Key Findings:**
  - ❖ Identified key factors influencing loan transactions like Purpose, Interest rates, Loan status, Loan category, State and Year.
- **Analysis insights:**
  - ❖ In the year 2025 only a greater number of loans sanctioned and more default loans but 2023 shows high paid off rate. So, there might be a change in policies or strategies in collecting the loan across time. This needs to be corrected.
  - ❖ Default interest rate = 13.59, Paid off interest rate = 12.88
    Current Interest rate = 13.07, Late interest rate = 13.46
    The Interest rates clearly impact on the loan repayment. So, decreasing the interest rates may increase the loan closure %.
  - ❖ Below average loan amount shows high sum of monthly payment and loan terms compared to Average loan amount. Monthly payment amount and terms should be reduced based on the loan amount.
  - ❖ If the default count is rising, tightening credit score requirements may work specifically for those regions.
  - ❖ Lowering the maximum loan term for new applicants in those states will reduce long-term exposure.

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10. Conclusion

The integration of Excel and Power BI proved effective for end-to-end data analysis, from raw data to visual reporting.